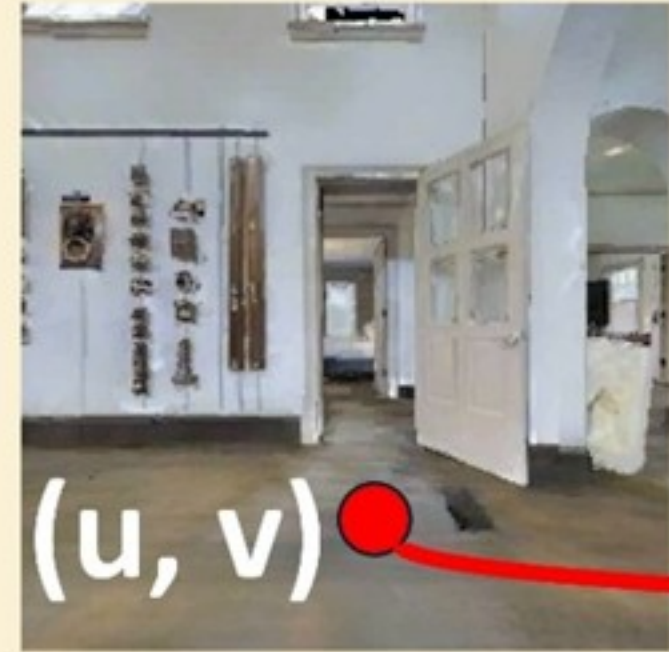


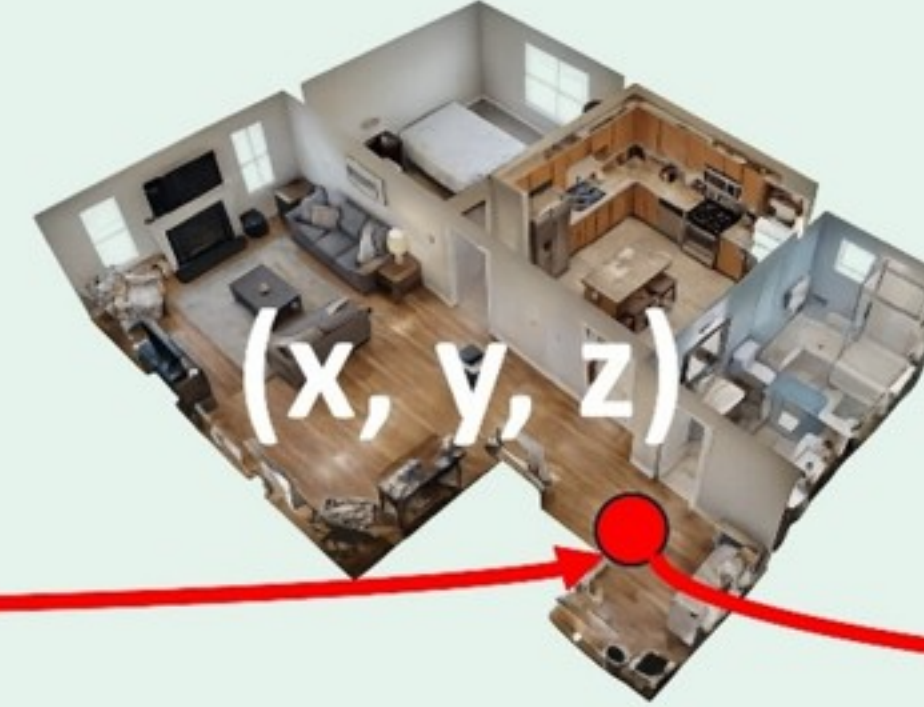
Execution Pipeline

Step 1. Predict Goal Pixel



Back-project 2D pixel
to 3D waypoint

Step 2. Project 3D Waypoint



Convert the waypoint
into low-level actions
via a local planner

Step 3. Low-Level Action Execution



Forward



Stop



Right



Left

Vision Model Backbone

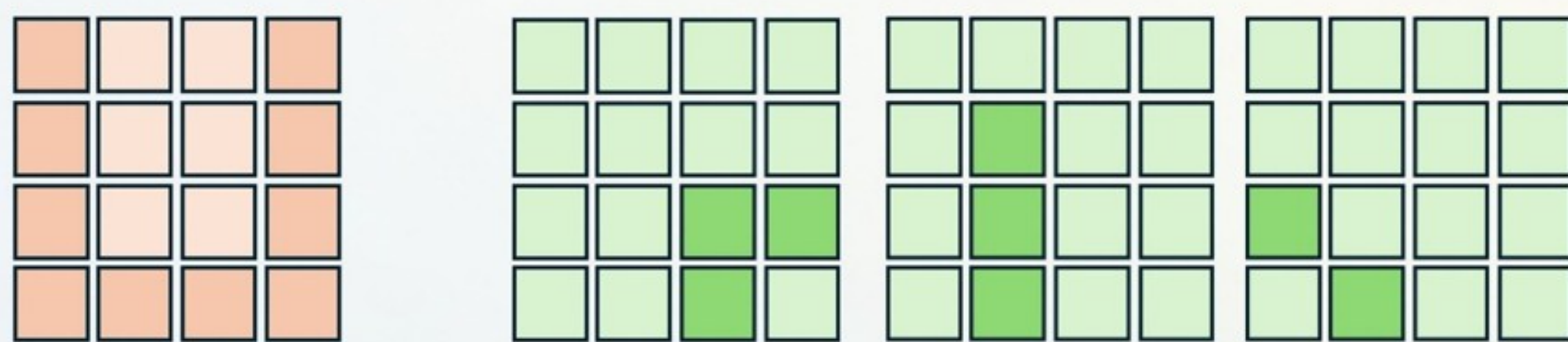
Padded Current
Observation Image

History Images (ViKeyMem)



Vision Encoder

Projection Layer



- Current Image Padding Token + $e_{padding}$
- Current Image RGB token
- History Trajectory Token + $e_{trajectory}$
- History RGB token

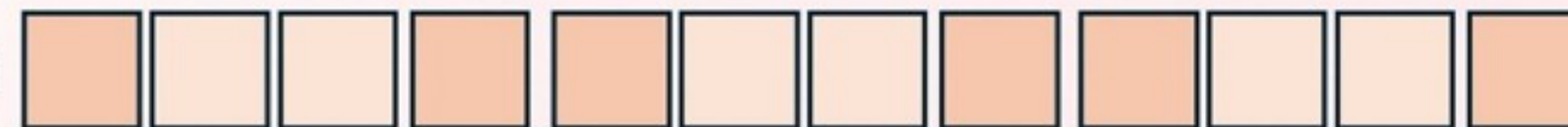


Large Language Model Backbone

You are an indoor navigation agent following..... <Task Introduction>

Instruction: <Current Episode Instruction>

Input:

- Current Step egocentric RGB image** , where the left/right/bottom gray padding indicates left/right turns and stop.
- History Images:** Previous <N> egocentric RGB keyframes  (latest), ,  (oldest).

Output: a pixel coordinate pair in the format XXX, YYY\n"

Vision-Language Model (InternVL3.0)

Training Objectives:

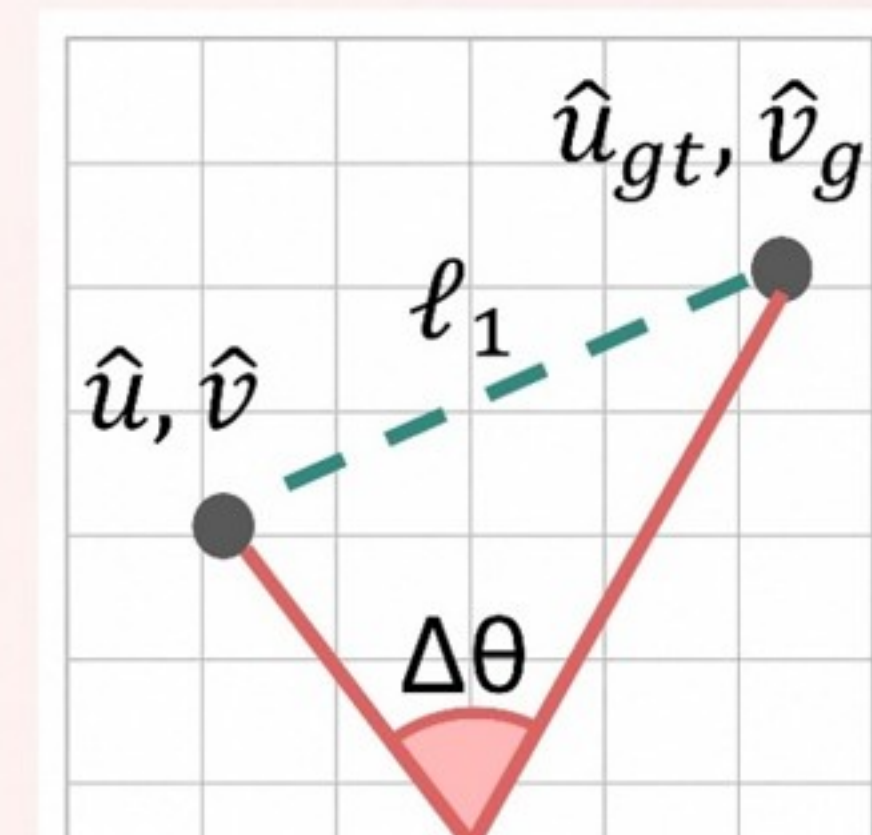
$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{Numeric} + \mathcal{L}_{Angular}$$

X_1 X_2 X_3 Y_1 Y_2 Y_3

Soft Predicted Pixel Coordinate

$\hat{u}$

$\hat{v}$



$$\mathcal{L}_{CE} = -\sum_{t=1}^T \log p(y_t | y_{<t}, x)$$

$$\mathcal{L}_{Numeric} = ||(\hat{u}, \hat{v}) - (u_{gt}, v_{gt})||_1$$

$$\mathcal{L}_{Angular} = 1 - \cos(\Delta\theta)$$